Telescope. There was nothing anomalous. Shuttle astronauts took time off from the reproduction of fruit flies, fish, and newts to look at Jupiter through binoculars. They reported seeing nothing. The impact of the millennium was beginning to look very much like a fizzle.

Then there was a report from a ground-based optical telescope in La Palma in the Canary Islands, followed by announcements from a radiotelescope in Japan; from the European Southern Observatory in Chile; and from a University of Chicago instrument in the frigid wastelands of the South Pole. In Baltimore the young scientists crowding around the TV monitor—themselves monitored by the cameras of CNN—began to see something, and in exactly the right place on Jupiter. You could witness consternation turn into puzzlement, and then exultation. They cheered; they screamed; they jumped up and down. Smiles filled the room. They broke out the champagne. Here was a group of young American scientists—about a third of them, including the team leader, Heidi Hammel, women—and you could imagine youngsters all over the world thinking that it might be fun to be a scientist, that this might be a good daytime job, or even a means to spiritual fulfillment.

For many of the fragments, observers somewhere on Earth noticed the fireball rise so quickly and so high that it could be seen even though the impact site below it was still in Jovian darkness. Plumes ascended and then flattened into pancake-like forms. Spreading out from the point of impact we could see sound and gravity waves, and a patch of discoloration that for the largest fragments became as big as the Earth.

Slamming into Jupiter at 60 kilometers a second (130,000 miles an hour), the large fragments converted their kinetic energy partly into shock waves, partly into heat. The temperature in the fireball was estimated at thousands of degrees. Some of the fireballs and plumes were far brighter than all the rest of Jupiter put together.

What is the cause of the dark stains left after the impact? It might be stuff from the deep clouds of Jupiter—from the region to which ground-based observers cannot ordinarily see—that welled up and spread out. However, the fragments do not seem to have penetrated